

An experimental population of fruit flies increases according to the law of exponential growth. There were 100 flies after the second day of the experiment and 300 flies after the fourth day. Approximately how many flies were in the original population and how fast is the population growing at $t = 5$. Also sketch the graph of exponential function that represent the population at time t .

$$P = c e^{kt} \quad \text{--- (A)}$$

$$t=2 \quad P_1 = 100$$

$$t=4 \quad P_2 = 300$$

$$100 = c e^{2k} \quad \text{--- (1)}$$

$$300 = c e^{4k} \quad \text{--- (2)}$$

$$\textcircled{2} \div \textcircled{1}$$

$$\frac{300}{100} = \frac{c e^{4k}}{c e^{2k}}$$

$$3 = e^{2k}$$

$$\ln 3 = 2k$$

$$\boxed{\frac{1}{2} \ln 3 = k} = .55$$

$$\textcircled{1} \Rightarrow 100 = c e^{1.1}$$

$$c = \frac{e^{1.1}}{100} = 0.03$$

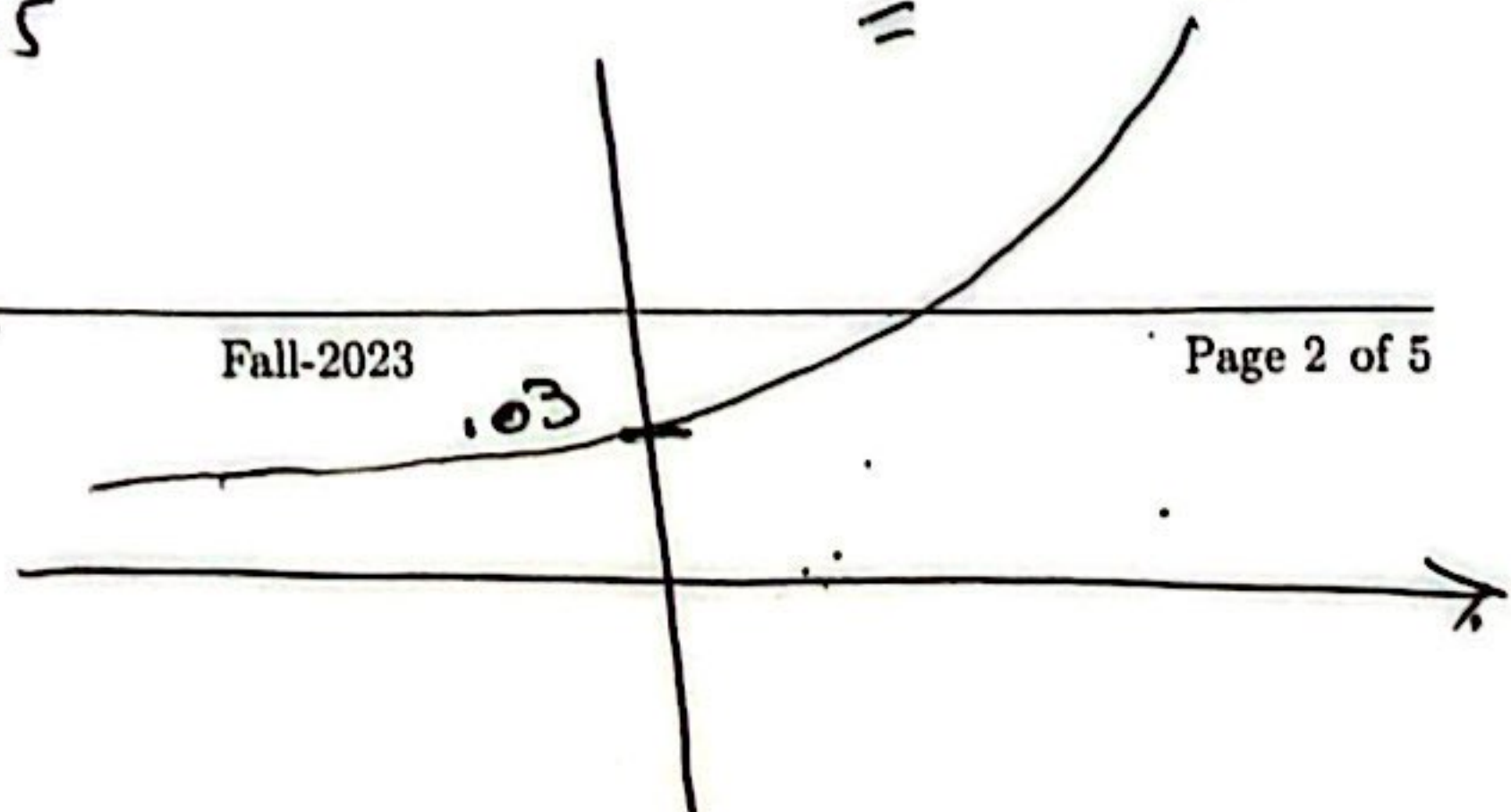
$$\textcircled{A} \Rightarrow \boxed{P = 0.03 e^{.55t}}$$

at $t=0 \quad P = 0.03$

initial population $.55 \times 5$

$$\left. \frac{dP}{dt} \right|_{t=5} = 0.03 \times .55 e^{.55t} = 0.0165 \times e^{.55 \times 5}$$

$$= .258$$



Verify that the given differential equation is non exact, also find its general solution.

$$ydx + (2xy - e^{-2y})dy = 0$$

$$f_n = y = M$$

$$f_g = 2xy - e^{-2y} = N$$

$$f_{ny} = 1$$

$$f_{yn} = 2y$$

$$\frac{f_{yn} - f_{ny}}{f_n} = \frac{2y - 1}{y} = \frac{2y}{y} - \frac{1}{y} = 2 - \frac{1}{y}$$

$$I.F. = e^{\int 2 - \frac{1}{y} dy} = \frac{e^{2y}}{y}$$

$$\frac{y e^{2y}}{y} \frac{d}{dy} (2xy - e^{-2y}) \frac{e^{2y}}{y} dy = 0$$

$$f_n = \frac{e^{2y}}{y}$$

$$f_g = 2xy + h(y), M = f = \frac{e^{2y}}{y} (2xy + h(y))$$

$$f_{ny} = \frac{2xy}{y} + h'(y) = 2x + h'(y), N = 2x e^{2y} - \frac{1}{y}$$

$$2x e^{2y} - \frac{1}{y} = 2x + h'(y)$$

$$\int M dx + \int N (term independent of x) dy = 0$$

$$x e^{2y} - \ln y = C$$

Identify the type of differential equation and find the solution of

$$(xy \cos(\frac{y}{x}) + y^2 \sin(\frac{y}{x}))dx + (x^2 \cos(\frac{y}{x}) - xy \sin(\frac{y}{x}))dy = 0.$$

$$\div x^2 \left[y \cos \frac{y}{x} + \left(\frac{y}{x}\right)^2 \sin\left(\frac{y}{x}\right) \right] dx + \left(\cos \frac{y}{x} - \frac{y}{x} \sin \frac{y}{x} \right) dy$$

$$(v \cos v + v^2 \sin v) dv + (\cos v - v \sin v)(x dv + v dx) = 0$$

$$v \cos v dx + v^2 \sin v dx + x \cos v dv + v \cos v dv - xv \sin v dv - v^2 \sin v dx = 0$$

$$2v \cos v dx + x \cos v dv + xv \sin v dv = 0$$

$$\cancel{2v \cos v dx} + \cancel{x \cos v dv} = - \cancel{xv \sin v dv}$$

$$\cancel{2v \cos v dx} = - \cancel{xv \sin v dv}$$

$$\cos v dv$$

$$2v \cos v dx = -x (\cos v + v \sin v) dv$$

$$-\frac{1}{x} dx = \frac{\cos v + v \sin v}{2v \cos v} dv$$

$$-\ln x = \frac{1}{2v} dv + \frac{1}{2} \frac{\sin v}{\cos v} dv$$

$$-\ln x = -\frac{1}{2} \ln v + \frac{1}{2} \ln \sec v + \ln c$$

$$\ln x^{-1} = \ln \frac{(\sec v)^{1/2} c}{v^{1/2}}$$

$$x = \frac{v^{1/2}}{c(\sec v)^{1/2}}$$

$$x = \frac{1}{c} \frac{(y/x)^{1/2}}{(\sec \frac{y}{x})^{1/2}}$$

Find the solution of initial value problem $y^{\frac{1}{2}} \frac{dy}{dx} - y^{\frac{3}{2}} = xe^x$; $y(0) = 4$

Let $y^{\frac{3}{2}} = u$

$\frac{3}{2} y^{\frac{1}{2}} \frac{dy}{dx} = \frac{du}{dx}$ Diff u.r.t x

$$\frac{2}{3} \frac{du}{dx} - u = xe^x$$

$$\frac{du}{dx} - \frac{3}{2} u = \frac{3}{2} xe^x \quad \text{--- (1)}$$

$$I.f = \exp\left[\int -\frac{3}{2} dx\right] = \exp\left[-\frac{3}{2}x\right] = e^{-\frac{3}{2}x}$$

$$e^{-\frac{3}{2}x} \frac{du}{dx} - \frac{3}{2} e^{-\frac{3}{2}x} u = \frac{3}{2} xe^x \cdot e^{-\frac{3}{2}x}$$

$$\frac{d}{dx} (u \cdot e^{-\frac{3}{2}x}) = \frac{3}{2} x e^{-\frac{1}{2}x}$$

Integrate

$$u e^{-\frac{3}{2}x} = \frac{3}{2} \left[x e^{-\frac{1}{2}x} - \int e^{-\frac{1}{2}x} (-2) dx \right]$$

$$y^{\frac{3}{2}} e^{-\frac{3}{2}x} = -3x e^{-\frac{1}{2}x} + 2 e^{-\frac{1}{2}x} + C$$

$$y^{\frac{3}{2}} e^{-\frac{3}{2}x} = \frac{-3x}{e^{\frac{1}{2}x}} - \frac{4}{e^{\frac{1}{2}x}} + C$$